

Voice-Based Concept Understanding Analyzer (VBCUA)

Project Description

The **Voice-Based Concept Understanding Analyzer (VBCUA)** is an AI-powered educational application that evaluates a student's conceptual understanding through voice responses. Instead of relying only on traditional written assessments, the system analyzes spoken explanations, converts speech into text using OpenAI Whisper, compares the response with reference concepts using semantic similarity, and generates an overall performance score.

The application also performs audio quality analysis by measuring speech duration, energy, speech rate, and pause percentage. Based on both speech characteristics and semantic understanding, the system calculates a final evaluation score and automatically generates a detailed PDF report for each student. The assessment results are also stored in a local database for future reference.

Features

- Upload student voice recordings (MP3, WAV, M4A)
- Automatic Speech-to-Text conversion using Whisper AI
- Semantic similarity evaluation using Sentence-BERT
- Audio quality analysis (duration, speech rate, energy, pauses)
- Intelligent scoring based on speech and conceptual understanding
- Automatic PDF report generation
- SQLite database for storing assessment results
- User-friendly web interface built with Streamlit

Technologies Used

Python, Streamlit, OpenAI Whisper, Sentence-Transformers (Sentence-BERT), Librosa, NumPy, SQLite, ReportLab, FFmpeg.

Applications

Educational institutions, online learning platforms, viva voce examinations, concept-based assessments, AI-assisted learning systems, and student performance evaluation.